IMDB Movie Review Sentiment Classifier

(Milestone 2: Data Understanding and Pipeline Setup)

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1.1 Milestone report

Project Overview:

The notebook prepares the IMDB movie-review dataset for a downstream binary sentiment-classification system. The task is to use a movie review as input and classify its sentiment as either positive or negative.

1. Project Definition

Project Title: IMDB Movie Review Sentiment Classifier

Problem Statement:

The objective of this project is to develop a machine learning system that automatically classifies movie reviews into positive or negative sentiment categories.

The project addresses the challenge of analysing large volumes of unstructured textual reviews manually. By applying Natural Language Processing (NLP) and machine learning techniques, the proposed system will identify sentiment expressed in movie reviews and provide an automated classification.

Intended Users and Application:

The system is intended for organisations and individuals interested in analysing customer opinions, including movie review platforms, entertainment businesses and researchers studying public opinion.

Machine Learning Task:

This project is a supervised binary text-classification problem.

Input: Movie review text.

Target: Sentiment.

Classes: Positive and negative.

Planned algorithms: Logistic Regression and Linear Support Vector Machine (SVM).

Planned text representation: Term Frequency-Inverse Document Frequency (TF-IDF).

Continuity from Milestone 1.

The project objective, dataset and classification approach remain consistent with Milestone 1. Milestone 2 extends the initial project by implementing data validation, exploratory data analysis, text preprocessing, cloud storage integration and a reproducible data preparation pipeline.

2. Data source, governance and ethics

The raw IMDB dataset is stored in Google Cloud Storage (GCS), which is used as the project's object-storage layer. The google colab notebook explicitly configures as,

PROJECT_ID = "dh-and-infra-for-ai",

BUCKET_NAME = "imdb-sentiment-project",

and RAW_GCS_PATH = "raw/IMDB Dataset.csv".

Data set Information:

Dataset	IMDB Movie Reviews
Size	50, 000 Reviews
format	CSV
Input	Review text
Target	Sentiment
Classes	Positive/Negative
Storage	Goolge Cloud Storage
GCS location	gs://imdb-sentiment-project/raw/IMDB Dataset.csv

Data Source:

Kaggle	https://www.kaggle.com/datasets/rehanliaqat17/imbd-dataset
Version	1
License	https://opendatacommons.org/licenses/pddl/1.0/
Privacy	The dataset contains movie review text rather than own customer records.
Bias	The dataset may not represent all movie audiences, languages, countries or demographic groups.
Class imbalance	dataset has 25,000 positive and 25,000 negative reviews
Data quality	Duplicate reviews may influence model evaluation if not handled correctly.
Misclassification	A sentiment classifier may misunderstand sarcasm, humour, mixed opinions or context.

3. Data description

Fig1:

```
Dataset shape: (50000, 3)
Columns: ['review', 'sentiment', 'split']
```

	review	sentiment
0	I caught this little gem totally by accident b...	positive
1	I can't believe that I let myself into this mo...	negative
2	*spoiler alert!* it just gets to me the nerve ...	negative
3	If there's one thing I've learnt from watching...	negative
4	I remember when this was in theaters, reviews ...	negative

Fig2:

```
...
Missing values:
review       0
sentiment    0
dtype: int64

Duplicate reviews: 418
```

Fig3:

```
...
Sentiment distribution:
sentiment
positive    25000
negative    25000
Name: count, dtype: int64

Sentiment proportions:
sentiment
positive    0.5
negative    0.5
Name: proportion, dtype: float64
```

Interpretation:

Dataset dimensions: Initial dataset contains 50,000 records.

Two relevant columns: review and sentiment.

Textual input data, Categorical target data.

Field	Data type	Description	Role

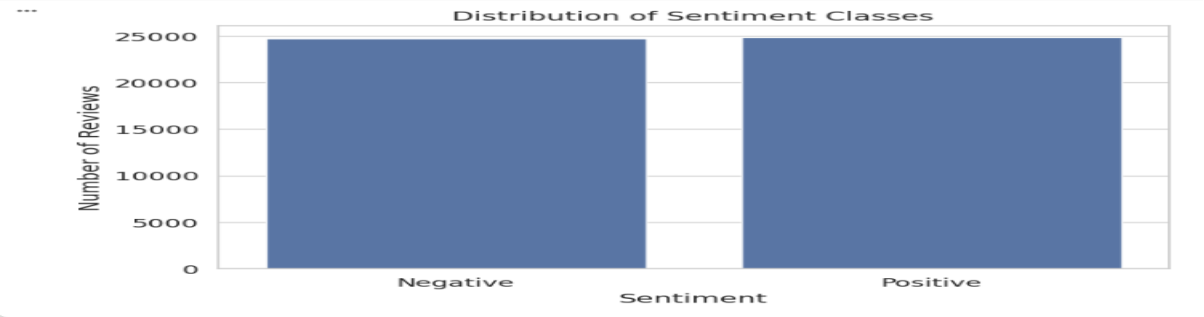
review	String	Movie review text	Input
sentiment	String/category	Positive or negative	Target
clean_review	String	Cleaned review text	Processed input
label	Integer	Positive = 1, negative = 0	Encoded target

Data quality results:

Quality check	Recorded result
Total records	50,000
Missing review values	0
Missing sentiment values	0
Duplicate reviews	418
Positive reviews	25,000
Negative reviews	25,000

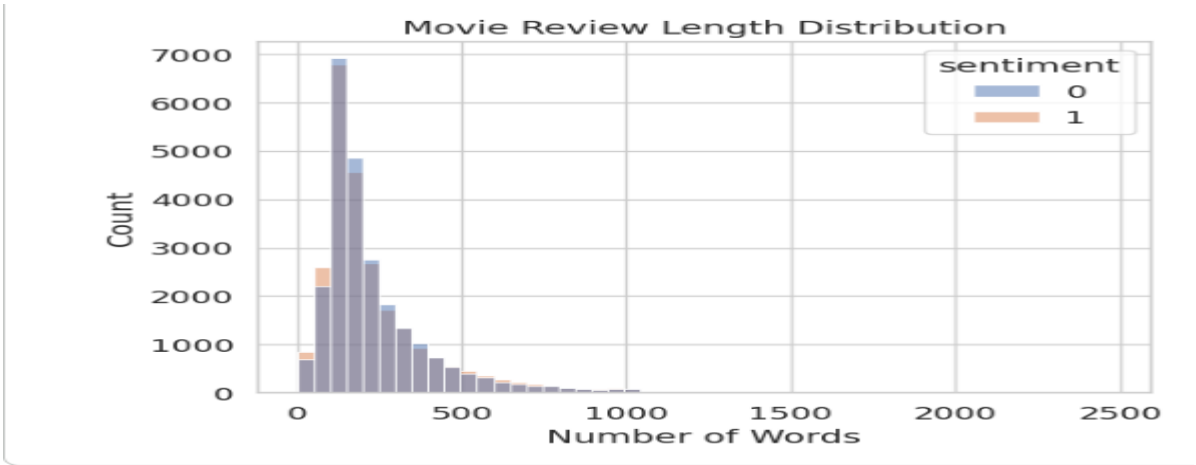
4. Exploratory data analysis

EDA - Distribution of sentiment classes:



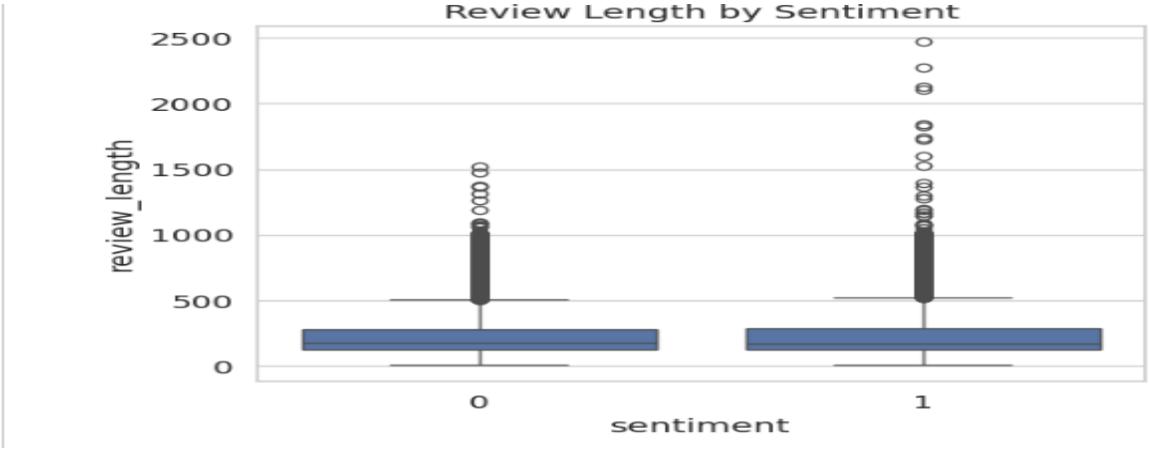
Interpretation: The dataset is balanced after removing duplicates, with a 0.4-point gap. There is no class-imbalance problem, so accuracy is a reliable metric, and no resampling or class weights are needed.

EDA - Movie review length distribution:



Interpretation: The distribution is strongly right skewed. Most reviews are short to medium, and a few are very long. The purple overlap region covers nearly everything, meaning the two classes have almost the same length profile.

EDA - Review length by sentiment (boxplot):



Interpretation: Length does not discriminate sentiment.

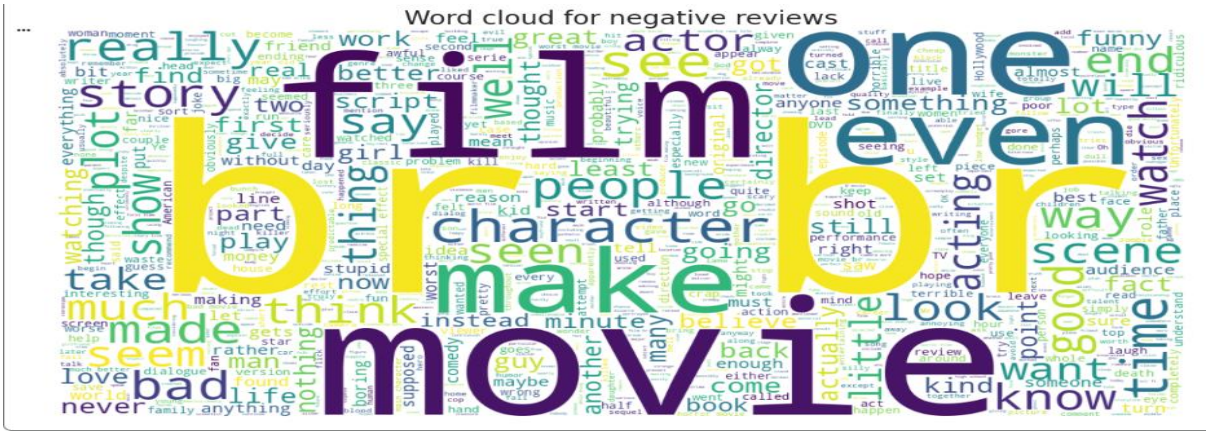
Outliers: There are many above ~500 words. Positive has the more extreme ones (up to ~2,470, versus ~1,520 for negative). Those few extreme positive reviews drag the mean slightly higher, 233 versus 230.

EDA - Word cloud: positive reviews



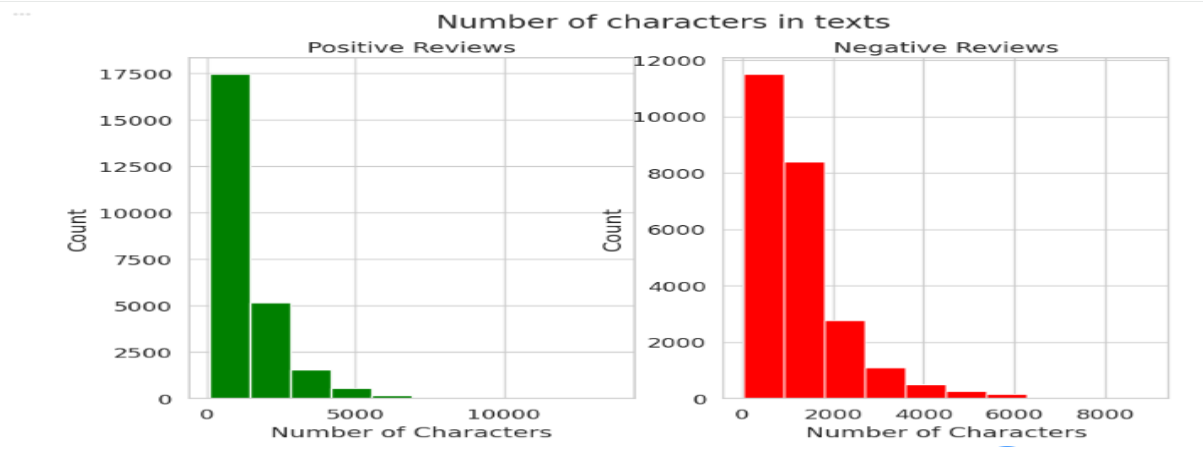
Interpretation: The biggest words are “br”, “the”, “of”, “in”, “is” and “a”, followed by phrases like “this movie” and “the film”.This cloud is not informative about positive sentiment.

EDA - Word cloud: negative reviews:



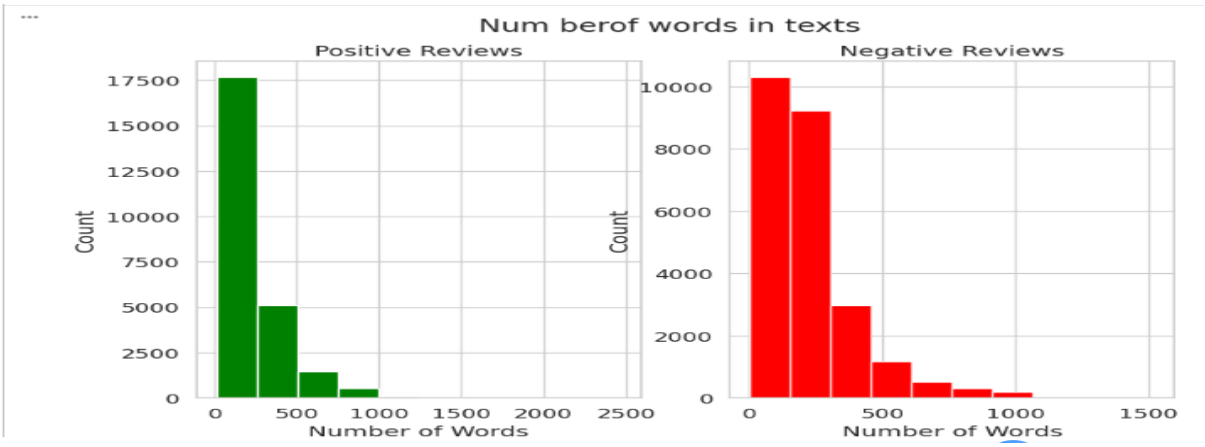
Interpretation: Word Cloud's default stop word list was applied here, so the function words are gone. This makes it look very different from above positive world could. Negative vocabulary is visible (“bad”, “worst”, “stupid”, “boring”, “waste”), but it is again overshadowed by generic words.

EDA - Number of characters in texts:



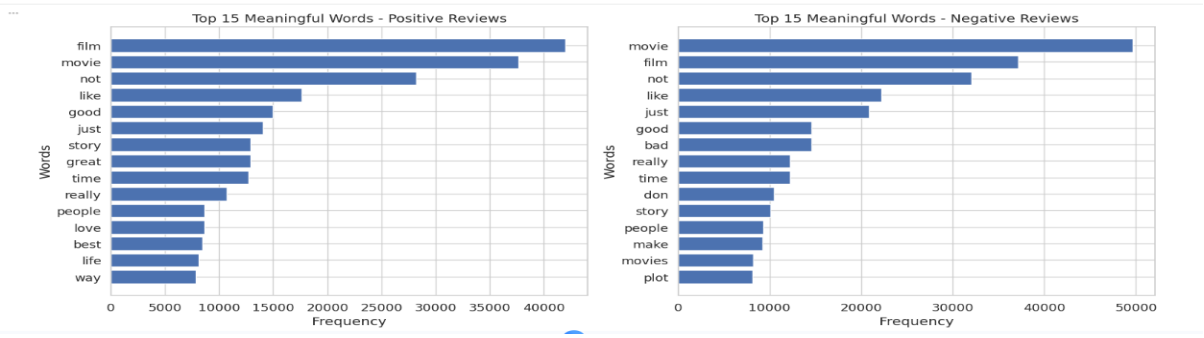
Interpretation: It repeats the length finding in character units. Both classes have the same shape: short reviews are common and very long ones are rare.

EDA - Number of words per review (two histograms):



Interpretation: It confirms plots 2 and 3, and it is largely redundant with them.

EDA - Top 15 meaningful words: positive vs negative:



Interpretation: About 11 of the 15 words are shared. The sentiment-bearing ones are: Positive: “great”, “love” and “best”. Negative: “bad”.

5. Design/ Architecture: Pipeline architecture and storage design:

The proposed architecture for the IMDB Movie Review Sentiment Classifier consists of a data acquisition, ingestion, validation, storage and processing pipeline. The system uses

Python and Google Colab for data processing and Google Cloud Storage (GCS) for persistent data storage.

The pipeline prepares the IMDB movie review dataset for binary sentiment classification, where each review is classified as either positive or negative. The prepared dataset will be used in the next stage for training and evaluating machine learning models.



Source: *https*://claude.ai/

6. Pipeline Implementation:

Pipeline implementation is presented with python code in google colab and available in github repository. This notebook runs from importing raw data from GCS buckets and perform descriptive and exploratory data analysis and prepares the data for Milestone 3.

https://github.com/vootlao/Data-Handling-and-Infrastructure-for-AI/blob/main/IMDB_Sentiment_Analysis_Milestone_2_EDA.ipynb